STATISTICAL FOUNDATIONS FOR MACHINE LEARNING

Notes/Scribes for TA Session - Week 2

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3. You flip a fair coin thrice and note the outcome (in $\{H,T\}^3$). Define the following events.

A is the event that all tosses have the same result (all H or all T).

B is the event that the you see at least one H and at least one T.

C is the event that you see at most one H.

(a) Are events A and B independent? **[1 mark]**

(b) Are events B and C independent? **[1 mark]**

(c) Are events A and C independent? **[1 mark]**

There are $2^3 = 8$ equally likely outcomes:

$$\Omega = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$$

First find the probabilities

Event A: all tosses have the same result

$$A = \{HHH, TTT\}$$

$$P(A) = \frac{2}{8} = \frac{1}{4}$$

Event B: at least one H and at least one T

$$B = \{HHT, HTH, HTT, THH, THT, TTH\}$$

$$P(B) = \frac{6}{8} = \frac{3}{4}$$

Event C: at most one H

$$C = \{TTT, HTT, THT, TTH\}$$

$$P(C) = \frac{4}{8} = \frac{1}{2}$$

For two events X, Y to be independent:

$$\boxed{P(X \cap Y) = P(X)P(Y)}$$

(a) Are A and B independent?

A and B cannot occur together:

$$A \cap B = \varnothing$$

Therefore,

$$\boxed{P(A \cap B) = 0}$$

But

$$P(A)P(B) = \frac{1}{4} \cdot \frac{3}{4} = \frac{3}{16} \neq 0$$

Hence,

No, A and B are not independent.

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(b) Are B and C independent?

$$B \cap C = \{HTT, THT, TTH\}$$

Thus,

$$P(B \cap C) = \frac{3}{8}$$

Now,

$$P(B)P(C) = \frac{3}{4} \cdot \frac{1}{2} = \frac{3}{8}$$

Since

$$P(B \cap C) = P(B)P(C)$$

$$\boxed{\text{Yes, B and C are independent.}}$$

(c) Are A and C independent?

$$A \cap C = \{TTT\}$$

Therefore,

$$P(A \cap C) = \frac{1}{8}$$

But

$$P(A)P(C) = \frac{1}{4} \cdot \frac{1}{2} = \frac{1}{8}$$

Hence,

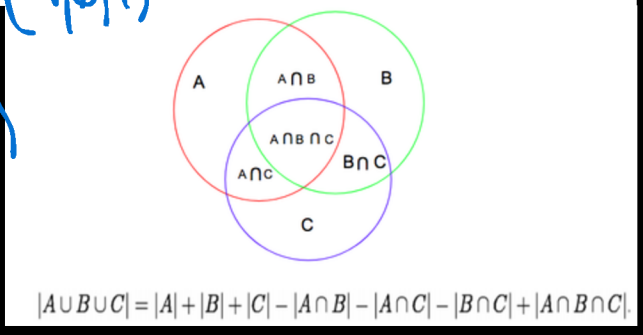
$$\boxed{\text{Yes, A and C are independent.}}$$

Final Answers

Part	Answer
(a) A, B	$\boxed{\text{Not independent}}$
(b) B, C	$\boxed{\text{Independent}}$
(c) A, C	$\boxed{\text{Independent}}$

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10. I choose a number uniformly in the range $[1, 1000000]$. Using inclusion exclusion principle, determine the probability that the chosen number is divisible by one or more of 4, 6, and 9.



Let

- A = number is divisible by 4
- B = number is divisible by 6
- C = number is divisible by 9

We want

$$P(A \cup B \cup C)$$

$$P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$$

1. Individual sets

$$|A| = \left\lfloor \frac{1,000,000}{4} \right\rfloor = 250,000$$

$$|B| = \left\lfloor \frac{1,000,000}{6} \right\rfloor = 166,666$$

$$|C| = \left\lfloor \frac{1,000,000}{9} \right\rfloor = 111,111$$

2. Pairwise intersections

A number divisible by both 4 and 6 must be divisible by

$$\operatorname{lcm}(4, 6) = 12$$

so

$$|A \cap B| = \left\lfloor \frac{1,000,000}{12} \right\rfloor = 83,333$$

Similarly,

$$\operatorname{lcm}(4, 9) = 36$$

$$|A \cap C| = \left\lfloor \frac{1,000,000}{36} \right\rfloor = 27,777$$

and

$$\operatorname{lcm}(6, 9) = 18$$

$$|B \cap C| = \left\lfloor \frac{1,000,000}{18} \right\rfloor = 55,555$$

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2. Pairwise intersections

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and

$$\text{lcm}(6,9) = 18$$

$$|B \cap C| = \left\lfloor \frac{1,000,000}{18} \right\rfloor = 55,555$$

3. Triple intersection

A number divisible by all $4, 6, 9$ must be divisible by

$$\text{lcm}(4,6,9) = 36$$

Therefore,

$$|A \cap B \cap C| = \left\lfloor \frac{1,000,000}{36} \right\rfloor = 27,777$$

4. Apply inclusion-exclusion

$$|A \cup B \cup C| = 250000 + 166666 + 111111 - 83333 - 27777 - 55555 + 27777$$

$$= 388,889$$

Therefore,

$$P(A \cup B \cup C) = \frac{388889}{1000000}$$

$$\boxed{P(A \cup B \cup C) = 0.388889}$$

or approximately

$$\boxed{38.89\%}$$

Answer: $\boxed{38.8889\%}$

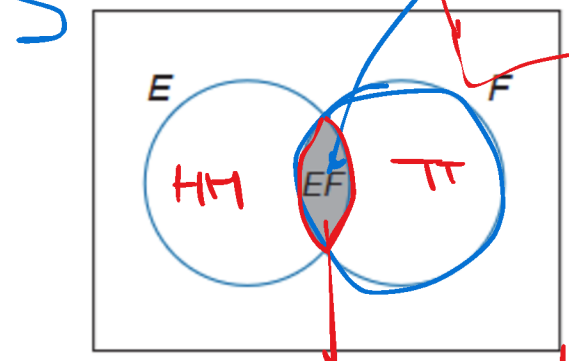
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Conditional Probability

Conditional probability of E given that F has occurred, and is denoted by

$P(E|F)$

$= \frac{P(E \cap F)}{P(F)} = \frac{2}{3}$



$P(F) = 3/4$

$P(E \cap F) = P(HT, TH) = 2/4$

A general formula for $P(E|F)$ that is valid for all events E and F. If the event F occurs, then in order for E to occur it is necessary that the actual occurrence be a point in both E and F; that is, it must be in EF. Now, since we know that F has occurred, it follows that F becomes our new (reduced) sample space and hence the probability that the event EF occurs will equal the probability of EF relative to the probability of F. That is,

$$P(E|F) = \frac{P(EF)}{P(F)}$$

if E and F are independent,

$P(E).P(F) = P(EF)$

$$P(E|F) = \frac{P(EF)}{P(F)}$$

$$P(E|F) = \frac{P(E).P(E)}{P(F)}$$

$P(E|F) = P(E)$

Similarly,

$P(F|E) = P(F)$

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Conditional Probability

$$P(E|F) = \frac{P(EF)}{P(F)}$$

$$P(F|E) = \frac{P(EF)}{P(E)}$$

$$P(F \mid E) = \frac{P(E \mid F)P(F)}{P(E)}$$

Standard Bayes Formula

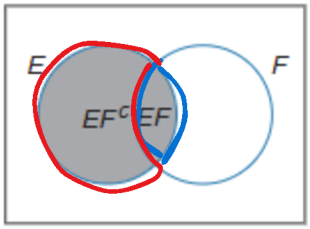
Bayes Formula

Let E and F be events. We may express E as

$$E = EF \cup EF^c$$

As EF and EF^c are clearly mutually exclusive, we have by Axiom 3 that

$$\begin{aligned} P(E) &= P(EF) + P(EF^c) \\ &= P(E|F)P(F) + P(E|F^c)P(F^c) \\ &= P(E|F)P(F) + P(E|F^c)[1 - P(F)] \end{aligned}$$



LAW OF TOTAL PROBABILITY

Equation states that the probability of the event E is a weighted average of the conditional probability of E given that F has occurred and the conditional probability of E given that F has not occurred: Each conditional probability is given as much weight as the event it is conditioned on has of occurring.

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